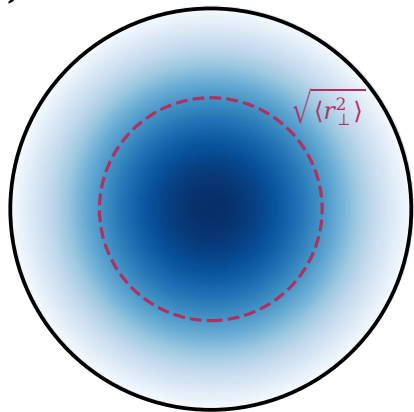


(a)

$$R_0 = 0.81 \text{ fm}$$

$$E_{\parallel} \propto J_0(x_{0,1} r/R_0)$$

(b)